

Amrit Gopinath

Undergraduate Researcher • Natural Language Processing • Language-Model Interpretability

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EDUCATION

Sri Sivasubramaniya Nadar College of Engineering, Chennai

2024–

Present

B.E. Computer Science and Engineering | CGPA: 9.383/10.0 | Department Rank: 1st

RESEARCH INTERESTS

Multilingual NLP | Representation Learning | Language-Model Interpretability | Mixture-of-Experts Models | Model Introspection | Classical and Low-Resource Tamil NLP

RESEARCH PAPERS

A Declarative–Procedural Perspective on Expert Routing in Bilingual MoE Language Models

2026

ARR Submission | arXiv Preprint | First Author

- Studied whether expert routing in bilingual English–German mixture-of-experts language models reflects language identity, lexical information, or higher-level grammatical and syntactic structure.
- Built controlled bilingual corpora, curriculum and no-curriculum training schedules, and held-out lexical, grammatical, and syntactic probe sets.
- Analyzed routing behaviour using entropy, mutual information, conditional mutual information, expert-use distributions, Jensen–Shannon divergence, frequency-controlled analysis, and targeted routing ablations.
- Led the research design, experimental pipeline, analysis, and manuscript preparation.

Exploring Swin Transformers, Luminance Input, and Hybrid Architectures for Synthetic Image Detection

2026

MediaEval 2026 | Working Notes | First Author

- Evaluated Swin Transformer, luminance-only, CNN–sequence, and hybrid architectures for distinguishing synthetic from authentic imagery.
- Best-performing Swin-based system achieved **0.6997 F1**; investigated whether luminance information and localized patch evidence improve robustness across synthetic-image generators.

FinMMEval 2026 Shared Task Submissions – TextSentinels

2026

CLEF 2026 | Working Notes – Accepted | Co-author

- Multilingual Financial MCQA:** developed FAISS-based retrieval with model routing and scoring across English, Chinese, and Arabic; ranked **6th**, **6th**, and **7th**, respectively.
- Financial QA:** built a BM25-grounded generation pipeline over SEC filings and tabular financial disclosures, improving local ROUGE-1 F1 from **0.04** to **0.28**.
- Financial Decision Making:** developed an iterative trading pipeline progressing from FinBERT sentiment features to XGBoost with sentiment memory; achieved **1.14 cumulative return** and **2.03 Sharpe ratio** in forward testing.

SELECTED RESEARCH

Having a State Is Not Knowing It

2026

Apart Research Sprint | Research Submission | Co-author

- Investigated whether language models can introspect on causally relevant internal decision states, separating possession of an internal state from the ability to verbally identify or predict it.
- Designed a hierarchy of falsifiable introspection capabilities spanning native self-report, trainable self-report, external decoding, counterfactual direction prediction, and intervention-magnitude prediction.
- Replicated concept-injection experiments showing that internal interventions can causally steer downstream behaviour while remaining poorly accessible through ordinary verbal self-report.
- Built a controlled Qwen decision-state benchmark using measured logit-margin shifts as ground truth. Native

- self-report reached **0.24 F1**, trainable self-report reached **0.52 F1**, while external probes decoded the relevant state perfectly.
- Models predicted the *direction* of counterfactual interventions at **0.70 F1**, but failed to reliably predict intervention magnitude, separating qualitative state sensitivity from quantitative self-modeling.

RESEARCH EXPERIENCE

NLP Research Intern

2026

National Institute of Technology Tiruchirappalli | Manuscript in preparation for ICON 2026

- Conducted research on computational modeling of Classical Tamil *urai*, with an emphasis on low-resource generation, linguistic structure, and representation learning.
- Developed mBART-style encoder–decoder baselines and decoder-only evaluation pipelines for generating and evaluating commentary from source verses.
- Investigated syntax-aware structural tags and explicit linguistic features as inductive biases for low-resource Tamil generation.
- Evaluated out-of-domain generalization by applying trained models to Thirukural couplets.
- Studied clause structure, predicates, finite and non-finite constructions, and morphosyntactic information as potential signals for retrieval and representation learning.

TECHNICAL SKILLS

ML / NLP	PyTorch, Hugging Face Transformers, scikit-learn, FAISS, Stanza
Research Methods	Representation probing, MoE routing analysis, mutual information, counterfactual interventions, ablation studies, error analysis
Programming / Tooling	Python, C, Java, TypeScript, Git, Linux, LaTeX

SELECTED ACHIEVEMENTS

- **FinMMEval 2026**: ranked **6th** in English, **6th** in Chinese, and **7th** in Arabic.
- **Merit Scholarship**: 3rd Position, First Year.
- **Top 10**: GDG Hackathon.

ACADEMIC SERVICE

Reviewer / Sub-reviewer

2026

CLEF 2026 Working Notes

- Reviewed working-note submissions for methodological clarity, experimental validity, presentation quality, and reproducibility.